

JASON (ZHANHAO) ZHAO

Los Angeles, CA

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Education

University of Southern California (USC)

Los Angeles, CA

Bachelor of Science in Computer Science

Expected May 2027

- **Relevant Coursework:** Advanced Analysis of Algorithms, Data Structures, Computer Systems, Probability Theory
- **Honors:** Viterbi Dean's List (Fall 23, Spring 24, Spring 25)

Technical Skills

Languages: C++ (C++17/20), Python, Java, TypeScript, SQL

Core Concepts: Data Structures & Algorithms (STL), Multithreading, Low Latency Systems, Network Programming

Tools & Frameworks: Linux/Unix, Git, Docker, Node.js, Next.js, React, PostgreSQL, WebSockets

Competitive Programming & Engineering Leadership

USC ICPC Team (International Collegiate Programming Contest)

Aug 2023 – Present

Team Member

Los Angeles, CA

- Selected to represent USC in the **ICPC Southern California Regionals** for 3 consecutive years (2023, 2024, 2025) through rigorous internal qualifiers.
- Consistently solved **5+** algorithmic problems per contest for 2 years.
- Specialized in solving complex algorithmic problems (Graph Theory, Dynamic Programming) under strict time constraints using **C++ STL** and custom data structures.

MAKERX Technology & FRC Team 6399

Dec 2020 – Present

Co-Founder & Lead Technical Mentor

Remote / Los Angeles, CA

- **Startup Foundation:** Co-founded a STEM education startup to commercialize the robotics team's training infrastructure; negotiated partnerships and led the digital transformation of local FRC chapters.
- **Robotics Control:** Previously served as Lead Programmer, developing a robot control system based on a **Swerve Drive** kinematics model (Java/PID) that reduced task completion time by 25%.
- Mentored students on software best practices, version control (Git), and system debugging via the Viterbi Experience Program.

Project Experience

Real-time Collaborative Whiteboard | *Next.js, WebSocket, CRDT*

Oct 2025

- Engineered a real-time collaboration platform supporting 50+ concurrent users with **<50ms latency** synchronization.
- Implemented **CRDTs (Conflict-free Replicated Data Types)** using Yjs to handle concurrent edits and network partitioning, ensuring eventual consistency across distributed clients.
- Architected a scalable WebSocket server (Node.js) to manage persistent full-duplex connections and optimized payload size.
- Built a responsive UI with Next.js 15 and React 19, utilizing memoization to maintain high frame rates during complex drawing.

High-Performance Limit Order Book | *C++20, Linux, CMake, Lock-Free Concurrency*

Dec 2025

- Engineered a deterministic, single-threaded matching engine implementing **Price-Time Priority** (FIFO), achieving a benchmarked throughput of **19.2 million orders/second** on commodity hardware.
- Designed a custom **Slab Allocator** to enforce a "Zero Allocation" policy on the hot path, eliminating runtime **new/delete** overhead and preventing memory fragmentation.
- Implemented a **Wait-Free SPSC Ring Buffer** using C++20 atomics with explicit memory ordering to ensure thread-safe, low-latency communication between network ingestion and matching cores.
- Optimized data structures for **L1 Cache Locality** by utilizing Flat Maps and **Intrusive Linked Lists**, reducing pointer chasing and ensuring $O(1)$ complexity for order execution and cancellation.
- Mitigated **False Sharing** by enforcing cache-line alignment (64-byte padding) on atomic indices, and utilized branch prediction hints (`[[likely]]`) to minimize pipeline flushes in the critical loop.